

MM-Verify: Enhancing Multimodal Reasoning with Chain-of-Thought Verification

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Abstract

According to the Test-Time Scaling, the integration of External Slow-Thinking with the Verify mechanism has been demonstrated to enhance multi-round reasoning in large language models (LLMs). However, in the multimodal (MM) domain, there is still a lack of a strong MM-Verifier. In this paper, we introduce MM-Verifier and MM-Reasoner to enhance multimodal reasoning through longer inference and more robust verification. First, we propose a two-step MM verification data synthesis method, which combines a simulation-based tree search with verification and uses rejection sampling to generate high-quality Chain-of-Thought (COT) data. This data is then used to fine-tune the verification model, MM-Verifier. Additionally, we present a more efficient method for synthesizing MMCOT data, bridging the gap between text-based and multimodal reasoning. The synthesized data is used to fine-tune MM-Reasoner. Our MM-Verifier outperforms all larger models on the MathCheck, MathVista, and MathVerse benchmarks. Moreover, MM-Reasoner demonstrates strong effectiveness and scalability, with performance improving as data size increases. Finally, our approach achieves strong performance when combining MM-Reasoner and MM-Verifier, reaching an accuracy of 65.3 on MathVista, surpassing GPT-4o (63.8) with 12 rollouts. Our code is made available <https://github.com/Aurora-slz/MM-Verify>.

1 Introduction

Large language models (LLMs) have demonstrated exceptional performance across diverse tasks spanning myriad domains (OpenAI, 2023a; Touvron et al., 2023). Based on LLMs, MLLMs (Zhao et al., 2023; Wu et al., 2023; Bai et al., 2024) also show strong understanding ability among different modalities (Liu et al., 2023b; Bai et al.,

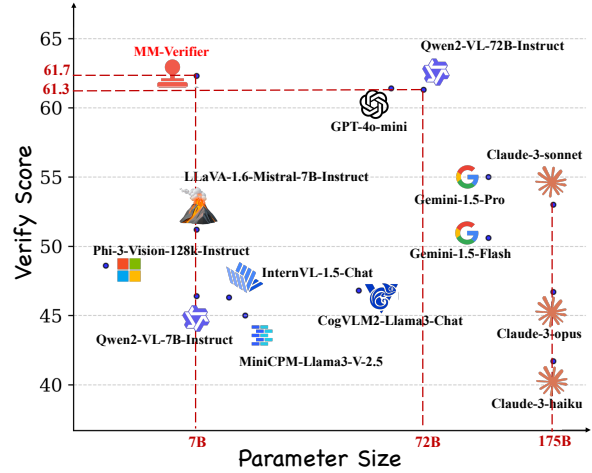


Figure 1: Our 7B MM-Verifier outperform all other models, even large models like GPT-4o, Gemini and Claude on the MathCheck Outcome-Judging benchmark.

2023b). They have demonstrated strong performance in image classification (Chen et al., 2024b), image understanding (Li et al., 2023b,c), image captioning (Bai et al., 2023b), visual question answering (Liu et al., 2023b,a) and image-text retrieval (Chen et al., 2024b). Recently, MLLMs have also made significant strides in solving mathematical problems (Liang et al., 2023; Huang et al., 2024). Researcher in our community have made efforts in designing strong reasoning models and algorithms (Thawakar et al., 2025; Du et al., 2025). Despite the effort made in processing MLLMs, we still face two challenges:

Lack of Strong MM Verifiers In pure-text LLMs, models can improve through self-critic methods (Weng et al., 2022; Sun et al., 2024). However, as shown in Table 6, such methods face challenges in enhancing performance in multimodal models. Therefore, it is crucial to develop robust multimodal (MM) verifiers.

Lack of Long COT Reasoning Data In the pure-text domain, models like DeepSeek-R1 (Guo

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et al., 2025), s1 (Muennighoff et al., 2025), and LIMO (Ye et al., 2025) have demonstrated the effectiveness of Long COT data. However, in the multimodal domain, most collected mathematics problems are not in the Long COT format (Li et al., 2024a; Lu et al., 2022; Chen et al., 2024a; Zhang et al., 2023; Shao et al., 2024; Cheng et al., 2025). Therefore, the development of Long COT synthetic methods is necessary to enhance the reasoning ability of MLLMs.

To address these issues, in this paper, we introduce two novel data synthesis methods and subsequently train MM-Verifier and MM-Reasoner. First, we perform a tree search, using simulations as rewards, to generate high-quality long MMCOT data. Next, we fine-tune MLLMs on our data, resulting in long chain-of-thought responses. These data are then Given that MLLMs generate long COT responses, we then apply rejection sampling to further enhance their verification capabilities, leading to the proposal of MM-Verifier. After the introduction of MM-Verifier, we observed that long COT data significantly improves model reasoning performance. As a result, we aim to synthesize large amounts of long COT data to enhance the performance of base MLLMs. Since tree search can be computationally expensive, we use the MAVIS dataset (Zhang et al., 2024e), leveraging the descriptions of patterns in MAVIS and inputting them into the pure-text reasoning model, Qwen QWQ. We then pair these patterns with the corresponding responses from the QWQ model. This method has proven to be effective, scalable, and capable of efficiently constructing large amounts of long MMCOT data.

The core contributions are summarized as follows:

- **MM Reasoning Data Synthesis Method** We propose two novel data synthesis methods for both our MM-Verifier and MM-Reasoner. First, we introduce a two-step MM-verification data synthesis approach that combines simulation-based tree search with GPT-4 verification and rejection sampling to generate high-quality COT data. Additionally, we use graphical software to link multimodal geometric shapes with textual descriptions, enabling the generation of multimodal reasoning data through a purely text-based reasoning model.
- **MM-Verifier** We introduce a new multimodal Outcome Reward Model (ORM) called MM-

Verifier. MM-Verifier achieves state-of-the-art (SOTA) performance on the MathCheck benchmark, surpassing closed-source models such as GPT-4, Gemini, and Claude. Furthermore, our MM-Verifier-7B outperforms Qwen2-VL-72B across all metrics on the MathVista and MathVerse benchmarks.

- **Scalability of MM-Reasoner** We propose a novel MM-Reasoning model based exclusively on our synthetic COT data. Although our MM-Reasoner does not achieve SOTA performance, it outperforms the baseline models and demonstrates scalability as the size of the training dataset increases. This provides new insights for the development of more powerful MM-Reasoners.
- **Strong Performance** By combining MM-Verifier and MM-Reasoning, with a model size of only 7B parameters, we outperform both GPT-4 and human performance on the MathVista benchmark, highlighting the strong performance of our method.

2 Related Work

2.1 MLLMs for Mathematics

Commonly Used MLLMs The integration of visual knowledge into LLMs has become a pivotal area of research due to the rapid advancements in LLMs. MLLMs combine vision information from vision encoders with LLMs, thus enabling these models to process and interpret visual inputs for various visual tasks (Liu et al., 2023c; Zhang et al., 2022; Li et al., 2022b) with enhanced accuracy and efficiency. Pioneering frameworks like CLIP (Radford et al., 2021) leverage contrastive learning on expansive image-caption datasets to align modalities, forming the groundwork for cross-modal comprehension. Various adapters (Liu et al., 2023b,a; Li et al., 2023b, 2022a; Jian et al., 2023; Lu et al., 2023a) are introduced to further integrate different modalities. For example, LLaVA (Liu et al., 2023b,a) employs a straightforward MLP to inject the vision information into LLMs. Whereas more complex implementations like the Q-Former in BLIP (Li et al., 2022a, 2023b) utilize cross-attention to enhance modality integration.

Recent studies (Wang et al., 2024b; Chen et al., 2023; Liu et al., 2023b,a; Li et al., 2023a; Zhang et al., 2024b; Zhuang et al., 2024; Luo

img +
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dataset

et al., 2025) aim to enhance MLLM performance by improving the quality of both pre-training and fine-tuning datasets. Models such as LLaVA (Liu et al., 2023b,a), ShareGPT4V (Chen et al., 2023), LLaVA-Next, LLaVA-OneVision (Liu et al., 2023b,a), Qwen2-VL, and Qwen2.5-VL (Bai et al., 2023b) have demonstrated significant advancements in understanding and executing complex instructions through instruction tuning. Leveraging large-scale training data, these models have also achieved strong performance in solving mathematical problems (Lu et al., 2023b).

MLLMs Designed for Math Problems In real-world applications, vision inputs are commonly used to present mathematical problems for models to solve. As a result, it is crucial for Vision-Language Large Models (MLLMs) to demonstrate strong mathematical capabilities. Meidani et al. (Meidani et al., 2023) pioneered the use of symbolic data to train a Vision-Language Model (VLM) with mathematical proficiency. Building on this work, UniMath (Liang et al., 2023) combined vision, table, and text encoders with LLMs, achieving state-of-the-art performance at the time. Additionally, Huang et al. (Huang et al., 2024) succeeded in solving algebraic problems that involved geometric diagrams.

Another noteworthy line of research involves using LLMs to tackle geometric problems. G-LLaVA (Gao et al., 2023) fine-tuned LLaVA (Liu et al., 2023b) with geometric data, reaching SOTA performance in geometry. Subsequently, MAVIS (Zhang et al., 2024e) and EAGLE (Li et al., 2024b) achieved SOTA results by introducing math-specific encoders and amassing large amounts of mathematical data.

2.2 LLM-as-a-Judge

In the Reinforcement Learning from Human Feedback (RLHF) or MCTS-based inference, Reward Models (RMs) are employed to assess and score the quality of model outputs, thereby guiding the optimization or reasoning path of LLMs (Chen et al., 2025). Reward models can be categorized into Process Reward Models (PRMs) and Outcome Reward Models (ORMs).

Outcome Reward Models. ORM evaluates only the final mathematical results without considering the solution process. For instance, Qwen2.5-Math-RM-72B (Zhang et al., 2025), released by the Qwen team, assigns a single score to each mathe-

matical response.

Process Reward Models. PRMs are more fine-grained, focusing on whether each step of the reasoning path is logical and correct, providing step-level feedback and guidance signals. For example, Math-Shepherd (Wang et al., 2024a) is trained on an automatically constructed (rather than manually annotated) process supervision dataset, scoring each step of mathematical reasoning. MATHMinos-PRM (Gao et al., 2024) introduces a novel two-stage training paradigm and incorporates step-wise natural language feedback labels. EurusPRM (Cui et al., 2025) utilizes implicit PRM, where ORM is trained to evaluate response-level labels. Qwen2.5-Math-PRM (Zhang et al., 2025), currently the SOTA PRM, proposes a consensus filtering mechanism combining Monte Carlo estimation and LLM-as-a-judge. Additionally, there are the Skywork-PRM series (o1 Team, 2024) and RLHFlow-PRM series (Xiong et al., 2024) models. Moreover, Liu et al. (2024) proposed Multimodal PRM based on Monte Carlo rollouts. For more comprehensive LLM-as-a-Judge please refer to the LLM-as-a-Judge survey (Gu et al., 2024).

3 Methodology

Table 1: Statistical details of our collected data.

Dataset	Subdataset	Number	Ratio
MM-Verify	Geometry3K	20226	33.84%
	FigureQA	10800	18.07%
	GEOS	882	1.48%
	Super-CLEVR	14446	24.17%
	TabMWP	13418	22.45%
	sum	59772	100%
MM-Reasoner	MAVIS-Geo	32146	100%

In this section, we introduce the construction process of MM-Verifier, as illustrated in Figure 2. Section 3.1 details the data synthesis methodology for MM-Verifier (Stage 1). Section 3.2 describes the data synthesis scheme employed in MM-Verifier (Stage 2). In Section 3.3, we explore the enhancement of multimodal model reasoning capabilities through the integration of long-COT data in pure text form.

3.1 Stage1: Long CoT MM-Verifier

3.1.1 Source Data Collection

MM-Verifier is designed to verify the correctness of a $\langle q, s \rangle$ pair by determining whether the

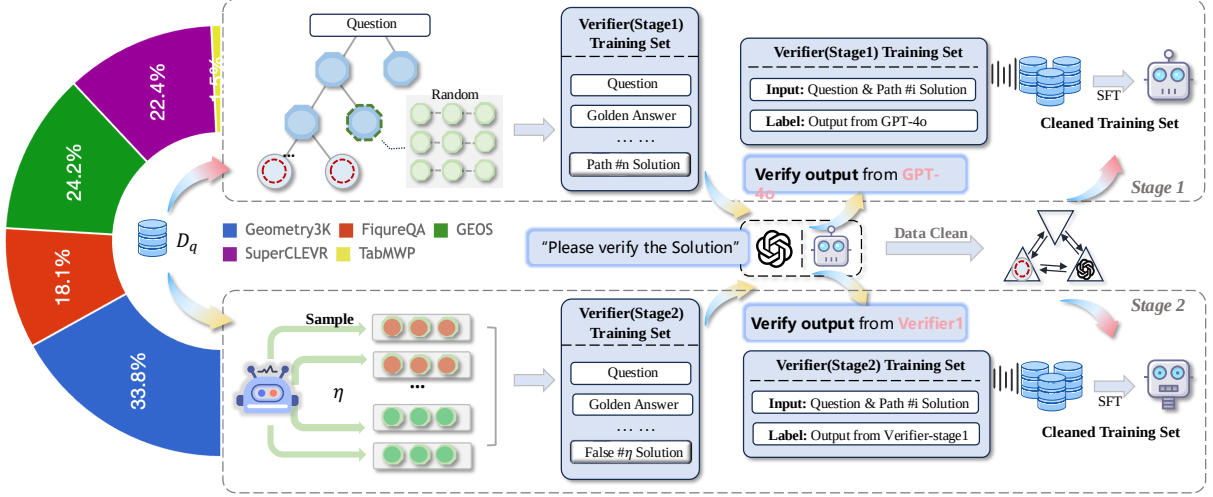


Figure 2: We present the pipeline for synthesizing MM-Verifier data. In Stage 1, we use a simulation-based algorithm for long-chain CoT reasoning and long verification. In Stage 2, we use the trained Verifier model from Stage 1 to further enhance it using rejection sampling, generating more long CoT verification data.

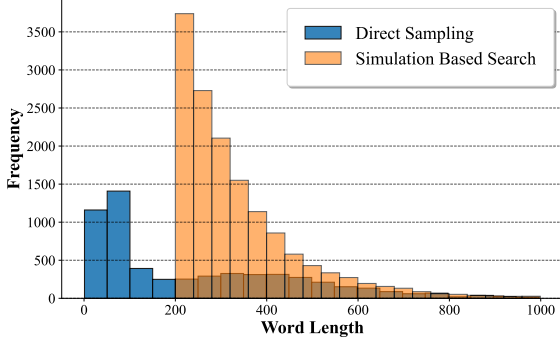


Figure 3: Answer length of direct sampling and simulated-based search. We can see the simulated-based search can synthesize longer COT answers.

solution s is correct. To achieve this goal, it is essential to synthesize diverse verification data from a variety of sources. Specifically, we construct the question source pool D_s from seven categories in MATH360V: Geometry3K, TabMWP, SuperCLEVR, UniGeo, FigureQA, and GEOS, with their statistical details summarized in Table 1. By collecting data from these categories, we obtained a diverse set of multimodal mathematics questions.

However, solutions in these datasets are typically very short, and many contain only answers. When verifying reasoning, the answer is often provided in a long CoT form. Therefore, we need to construct CoT data ranging from short to long, along with their corresponding verifications. To facilitate the generation of long CoT data and enhance the diversity of the training set, we design a simulation-based search algorithm to create extended reasoning trajectory data for training MM-Verifier.

3.1.2 Simulation-based Search Algorithm

Inspired by Monte Carlo Tree Search (MCTS), we propose a simulation-based search algorithm tailored for multimodal models. However, we do not directly apply the traditional MCTS tree search, as multimodal models often fail to generate reliable rewards, which results in suboptimal performance, as illustrated in Appendix A. To address this challenge, we introduce a simulation-based reward mechanism.

Starting from the root node q_i , we first simulate k child nodes for each node. For each child node, we perform simulations where the model directly generates answers based on the current node. For a tree node u_d , which represents a node at depth d , the ancestral path leading up to the root is denoted by the sequence $\{u_{d-1}, \dots, u_1\}$. The simulation answer for this path is given by:

$$\text{Simulation Answer} = LLM \left(\bigoplus_{i=1}^{d-1} u_i \right)$$

These simulations are repeated l times, and the correctness ratio of the *Simulation Answer* is used as the reward. Once the reward is obtained, we apply the MCTS algorithm for further simulation and data synthesis.

Using the simulation-based MCTS approach, for each question, we perform n rollouts and collect solution pairs $\langle q_i, p_j^i \rangle$, where $j \in \{1, 2, \dots, n\}$ and n represents the number of leaf nodes. These n solution pairs are then verified as positive and negative cases for training the MM-verifier.

Table 2: We compare performance of Qwen2-VL-Instruct-7B/72B, LLaMA-3.2-11B-Vision-Instruct with our MM-Reasoner on the MathVista testmini benchmark. GVQA: General VQA; MVQA: Math Target VQA.

Method	Qwen2-VL			LLaMA-3.2-11B-Vision			MM-Reasoner		
	ALL	GVQA	MVQA	ALL	GVQA	MVQA	ALL	GVQA	MVQA
<i>Sample 4</i>									
Majority Voting	57.1	65.7	49.8	45.2	50.8	40.4	59.4	65.2	54.3
Qwen2-VL-7B as Judgement	57.9	66.3	50.7	41.7	48.9	35.5	53.1	59.8	47.4
Qwen2-VL-72B as Judgement	53.4	60.4	47.4	46.3	49.8	43.3	53.7	59.8	48.5
MM-Verifier(Stage1)	58.8	64.1	54.3	50.0	56.3	44.6	60.6	66.5	53.7
MM-Verifier(Stage2)	59.8	67.0	53.7	50.4	55.9	45.7	61.5	65.2	58.3
<i>Sample 8</i>									
Majority Voting	61.1	68.5	54.8	48.3	52.8	44.4	62.2	68.0	57.2
Qwen2-VL-7B as Judgement	54.5	62.0	48.1	46.1	50.7	42.2	53.6	61.5	46.9
Qwen2-VL-72B as Judgement	56.2	62.4	50.9	46.2	51.1	42.0	53.9	61.1	47.8
MM-Verifier(Stage1)	61.6	66.5	57.4	51.4	56.3	47.2	63.8	68.7	59.6
MM-Verifier(Stage2)	62.5	68.5	57.4	52.1	57.6	47.4	65.3	69.8	61.5
<i>Sample 12</i>									
Majority Voting	62.9	68.5	58.1	51.3	56.5	46.9	64.8	68.7	61.5
Qwen2-VL-7B as Judgement	54.4	60.0	49.6	45.5	48.6	42.0	55.4	61.1	50.6
Qwen2-VL-72B as Judgement	55.7	61.1	51.1	46.5	49.8	43.7	55.6	61.5	48.7
MM-Verifier(Stage1)	63.7	70.2	58.1	55.0	59.8	50.9	64.3	69.1	60.1
MM-Verifier(Stage2)	64.1	70.4	58.7	55.9	60.9	51.7	65.2	69.8	61.3

3.1.3 Long COT Verification Data Synthesize

After obtaining the $\langle q_i, p_j^i \rangle$ pair, the next step is to verify p_j^i to determine whether q_i has been answered correctly. To do this, we use GPT-4o (gpt-4o-2024-08-06) to verify $\langle q_i, p_j^i \rangle$ using the instruction "Verify step by step..." (for the detailed prompt, see Figure 7). The model's output, denoted as v_i , serves as the target for the verifier. The resulting dataset collected at this stage is represented as D_v , which can be expressed as:

$$D_v = \{(q_i, p_j^i, v_i) \mid i = 1, \dots, m; j = 1, \dots, n\}$$

Additionally, we implement a data-cleaning strategy to filter high-quality synthetic data for training the MM-Verifier. For each data instance (q_i, p_j^i, v_i) in D_v , we first design an answer extraction prompt, as shown in Figure 9) and use LLaMA-3.2-3B-Instruct for answer extraction, denoted by $\text{extract}()$ for clarity. We then apply the following criteria for data cleaning:

- **Condition 1:** If $\text{extract}(p_j^i)$ matches the golden label $\text{extract}(y_i)$, and the final result of v_i is the answer is correct.
- **Condition 2:** If $\text{extract}(p_j^i)$ does not match $\text{extract}(y_i)$, and the final result of v_i is the answer is not correct.

We collect the instances that meet the above conditions, (q_i, p_j^i, v_i) , into D_{clean} . Any instance that

does not satisfy these conditions is discarded. The dataset D_{clean} is then used for supervised fine-tuning (SFT) on Qwen2-VL-7B-Instruct, yielding the first-stage verifier, MM-Verifier (Stage 1).

3.2 Stage 2: Rejection Sampling Further Improves Verification

In Stage 1, we obtained a Verifier with strong long-chain Chain-of-Thought (CoT) reasoning capabilities. In Stage 2, our goal is to improve the efficiency of the data synthesis process, reduce API costs, and further enhance the Verifier's capabilities.

To achieve this, we first generate corresponding solutions based on a given set of questions. By leveraging the long CoT reasoning ability of the MM-Verifier from Stage 1, we can generate additional long CoT verification data.

The synthetic data is then cleaned using string matching against the correct answer. The filtered data is subsequently fed into the MM-Verifier (Stage 1) for further training, resulting in the enhanced MM-Verifier (Stage 2).

3.3 Bridging the Gap Between Text and MM

When applying MM-Verifier, the base model generates multiple reasoning paths for a given question. The MM-Verifier then evaluates these paths, distinguishing between correct and incorrect inferences. This process inherently requires the base model to produce at least one plausible correct reasoning path for the MM-Verifier to recognize

Table 3: We compare performance of Qwen2-VL-Instruct-7B/72B, LLaMA-3.2-11B-Vision-Instruct with our MM-Reasoner on the MathVerse testmini benchmark. VD: Vision Dominant; VI: Vision Intensive; TL: Text Lite.

Method	Qwen2-VL				llama-3.2-11B-Vision				MM Reasoner			
	ALL	VD	VI	TL	ALL	VD	VI	TL	ALL	VD	VI	TL
<i>Sample 4</i>												
Majority Voting	20.1	19.8	19.0	17.7	17.8	15.9	17.4	19.8	22.9	20.7	23.0	23.2
Qwen2-VL-7B as Judgement	20.6	22.3	20.8	19.0	20.6	19.3	21.1	21.2	22.6	21.4	21.4	24.6
Qwen2-VL-72B as Judgement	21.5	20.7	21.1	22.0	20.2	20.8	18.1	21.6	23.0	22.0	22.3	24.0
MM-Verifier(Stage1)	24.0	25.3	22.6	22.8	21.9	23.2	20.8	22.5	24.8	22.2	25.6	26.0
MM-Verifier(Stage2)	24.3	23.6	23.2	23.1	22.4	21.8	23.0	24.4	25.3	23.1	23.5	27.4
<i>Sample 8</i>												
Majority Voting	22.9	21.8	22.1	23.5	23.5	20.6	22.8	25.5	24.5	20.6	25.0	24.5
Qwen2-VL-7B as Judgement	21.1	20.6	20.7	21.1	21.1	19.7	21.1	20.0	21.7	19.2	24.6	20.3
Qwen2-VL-72B as Judgement	21.5	20.3	21.2	21.2	20.8	19.5	20.2	20.4	22.3	19.4	24.4	23.1
MM-Verifier(Stage1)	25.1	22.7	24.2	25.5	24.8	21.6	24.4	25.4	25.3	23.1	23.5	27.4
MM-Verifier(Stage2)	25.2	24.2	24.1	25.1	25.0	22.7	26.3	25.9	25.7	23.2	25.3	26.3
<i>Sample 12</i>												
Majority Voting	24.8	19.9	24.7	25.6	24.0	19.7	24.5	26.1	25.9	23.1	26.1	27.3
Qwen2-VL-7B as Judgement	22.0	20.1	22.0	22.5	20.0	18.9	19.9	23.1	21.8	19.5	22.7	23.4
Qwen2-VL-72B as Judgement	22.3	20.4	22.3	23.0	20.7	19.4	20.9	20.7	22.1	20.1	23.2	25.0
MM-Verifier(Stage1)	25.7	23.1	26.5	26.5	24.5	21.1	23.2	25.5	27.0	23.5	27.0	29.4
MM-Verifier(Stage2)	25.8	21.7	25.4	28.2	24.7	21.2	24.7	27.0	27.3	24.1	27.7	28.9

and validate. Therefore, in addition to a promising MM-Verifier, In this section, our goal is to synthesize long COT data to train a robust MM-Reasoner capable of learning long COT reasoning. However, synthesizing long COT data using tree search can be computationally expensive. Moreover, long COT pure text models have demonstrated strong performance. To efficiently generate long COT data, we propose distilling knowledge from pure text models to our MM-Reasoner.

Specifically, we select data from MAVIS-GEOMETRY (Zhang et al., 2024e), which includes geometric pattern drawings along with textual instructions. By combining the geometric textual instructions with the original questions, we can feed them into a pure text reasoning model. The outputs generated by the reasoning model, Qwen-QwQ (Team, 2024) with prompts in Figure 8, are then collected as target labels for the MM-Reasoner training dataset, denoted as D_r . To ensure the quality of the data, we filter out incorrect QwQ-generated reasoning results from D_r . Then we use the filtered data to train Qwen2-VL-7B-Instruct with supervised fine-tuning (SFT), ultimately obtaining the MM-Reasoner model.

4 Experiments

4.1 Experiment Setting

Baseline Models. The base models for MM-Verifier and MM-Reasoner are Qwen2-VL-Instruct-7B (Bai et al., 2023b). For comparison, we include random selection and human performance as baselines, along with two types of closed-source models: GPT-4o (gpt-4o-2024-08-06) (OpenAI, 2023b) and Qwen-VL-Plus (Bai et al., 2023a). For open-source models, we evaluate ten MLLMs: mPLUG-Owl2-7B (Ye et al., 2024), MiniGPT4-7B (Zhu et al., 2023), LLaVA-1.5-13B (Liu et al., 2023a), SPHINX-V2-13B (Lin et al., 2023), Deepseek-VL (Lu et al., 2024), LLaVA-OneVision-7B (llava-onevision-qwen2-7b-ov-hf) (Li et al., 2024a), Qwen2-VL-Instruct-7B (Bai et al., 2023b), Llama-3.2-11B-Vision (Touvron et al., 2023), Math-LLaVA (Shi et al., 2024), and G-LLaVA-7B (Gao et al., 2023).

MM-Verifier Baselines. **Majority Voting:** Select the answer that appears most frequently among multiple candidate answers. **Qwen2-VL-7B and Qwen-VL-72B as Judgment:** We use Qwen2-VL-7B-Instruct and Qwen-VL-72B (Bai et al., 2023b) as the judgment model to assess the correctness of each candidate solution. If multiple candidates are deemed correct, we apply a majority voting

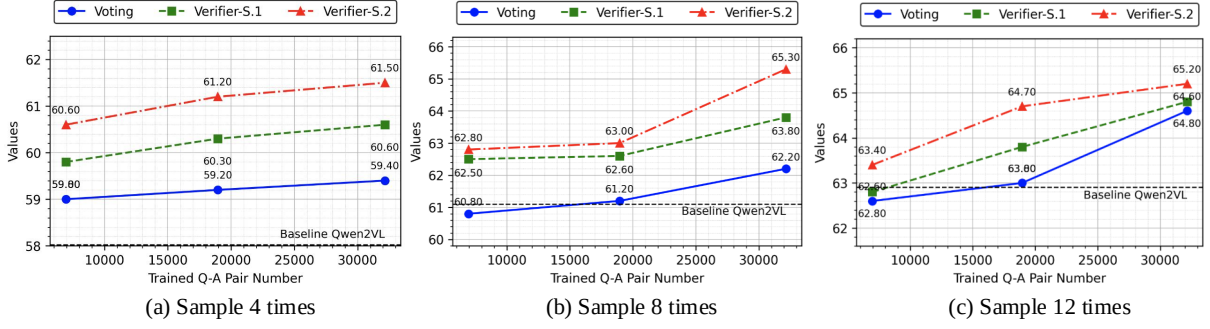


Figure 4: The performance of our MM-Reasoner can scale up using different MM-Verifiers. We can see with different scale MM-Reasoner the MM-Verifier consistently outperform majority voting and MM-Verifier Stage1.

mechanism to determine the final answer.

Benchmarks. We evaluate the performance of MM-Verifier, MM-Reasoner, and the baselines on two commonly used benchmarks: MathVista (Lu et al., 2023b) and MathVerse (Zhang et al., 2024d). Additionally, we assess the MM-Verifier on the Verify bench MathCheck (Multimodal Outcome Judging) (Zhou et al., 2024).

Settings. The settings include a maximum token limit of 4096, a top-k value of 5, a temperature of 0.3, and a repetition penalty of 1.05. All experiments are conducted on 8 NVIDIA H20 GPUs.

4.2 Effectiveness of MM-Verifier and MM-Reasoner

In this section, we demonstrate the effectiveness of both the MM-Verifier and MM-Reasoner. As shown in Figure 1, the MM-Verifier outperforms all other models in the MathCheck benchmark. Moreover, Tables 2 and 3 reveal that the MM-Verifier achieves strong performance on the MathVista and MathVerse benchmarks. Our results indicate that the MM-Verifier surpasses both the Majority Voting and Qwen2-VL-72B-Instruct methods. Specifically, MM-Verifier (Stage 2) delivers superior results, demonstrating its robust ability to verify answers and enhance model performance.

Furthermore, the MM-Reasoner outperforms powerful models, such as Qwen2-VL-Instruct-7B and LLaMA-3.2-11B-Vision, across all evaluation metrics. These findings clearly demonstrate that both the MM-Verifier and MM-Reasoner contribute significantly to performance improvements, underscoring their potential for addressing complex multimodal reasoning and verification tasks.

Interestingly, we observe that the performance of Qwen2-VL-72B, when used as a Judgment model, improves when verifying answers for Qwen2-VL

and LLaMA-3.2-11B-Vision (from sample 4 to sample 12). However, its performance drops when verifying MM-Reasoner outputs. This discrepancy arises because conventional models struggle to verify the correctness of longer outputs, while our MM-Verifier consistently maintains robust performance. This further emphasizes the versatility of the MM-Verifier across a wide range of scenarios.

4.3 Scalability of Our MM-Reasoner

The results in Figure 4 illustrate the scalability of MM-Reasoner with respect to the quantity of training data. As the amount of training data increases, the performance of the model consistently improves across all evaluation settings. Specifically, MM-Reasoner achieves steady performance gains when moving from 6,952 to 32,146 training samples, demonstrating its ability to effectively utilize larger datasets for better reasoning.

Additionally, both Verifier-S.1 and Verifier-S.2 show clear improvements as the training data grows, with Verifier-S.2 outperforming Verifier-S.1 in all cases. This trend highlights the effectiveness of the staged verification approach in enhancing reasoning accuracy.

These results emphasize the superiority of our method, as MM-Reasoner scales effectively with increasing training data, achieving higher performance and showcasing the robustness and adaptability of our approach.

4.4 MM-Verifier and MM-Reasoner achieved SOTA Performance

We leveraged our proposed MM-Reasoner and MM-Verifier together to enhance multimodal mathematical reasoning. Specifically, MM-Reasoner generated 12 diverse reasoning rollouts per query, while MM-Verifier systematically evaluated and filtered these rollouts, ensuring high-quality and

Table 4: We compare our MM-Verifier plus MM-Reasoner with closed and open source MLLMs, and other baselines. GPS: geometry problem solving; MWP: math word problem; FQA: figure question answering; TQA: textbook question answering; VQA: visual question answering.

Model	MATHVISTA						MATHVERSE				
	ALL	GPS	MWP	FQA	TQA	VQA	ALL	TD	TL	VI	VD
<i>Closed Source MLLMs & Other Baselines</i>											
Random	17.9	21.6	3.8	18.2	19.6	26.3	12.4	12.4	12.4	12.4	12.4
Human	60.3	48.4	73.0	59.7	63.2	55.9	64.9	71.2	70.9	61.4	68.3
GPT-4o	63.8	64.7	-	-	-	-	50.8	59.8	50.3	48.0	46.5
Qwen-VL-Plus	43.3	35.5	31.2	54.6	48.1	51.4	21.3	26.0	21.2	18.5	19.1
<i>Open-Source MLLMs</i>											
mPLUG-Owl2-7B	22.2	23.6	10.2	22.7	27.2	27.9	10.3	11.6	11.4	11.1	9.4
MiniGPT4-7B	23.1	26.0	13.4	18.6	30.4	30.2	12.2	12.3	12.9	12.5	14.8
LLaVA-1.5-13B	27.7	22.7	18.9	23.8	43.0	30.2	12.7	17.1	12.0	12.6	12.7
SPHINX-V2-13B	36.7	16.4	23.1	54.6	41.8	43.0	16.1	20.8	14.1	35.2	28.9
Deepseek-VL	34.9	28.4	55.9	26.8	32.9	34.6	19.3	23.0	23.2	20.2	18.4
LLaVA-OneVision-7B	51.1	66.8	54.8	39.4	57.6	40.8	20.7	25.5	21.8	20.9	21.2
Qwen2-VL-7B	52.5	44.7	59.7	55.4	52.5	49.7	20.1	23.7	16.3	19.0	19.8
llama-3.2-11B-Vision	46.5	38.0	51.6	50.9	51.9	39.7	17.8	19.2	17.9	15.6	15.5
G-LLaVA-7B	25.1	48.7	3.6	19.1	25.0	28.7	16.6	20.9	20.7	17.2	14.6
Math-LLaVA-13B	46.6	57.7	56.5	37.2	51.3	33.5	22.9	27.3	24.9	24.5	21.7
Ours	65.3	60.1	73.1	68.8	67.7	55.9	25.7	30.2	26.3	25.3	23.2

Input <question, solution>	
Q: As shown in the figure, the diameter CD of $\odot O$ crosses the midpoint G of chord EF, $\angle DCF = 20.0$, then $\angle EOD$ is equal to ()	
A: Let's think step by step.	
Given that CD is the diameter of circle O, and G is the midpoint of chord EF, thus OG is perpendicular to EF. Since $\angle DCF = 20^\circ$, and CD is the diameter, $\angle CED = 90^\circ$. Therefore, $\angle CGF = 90^\circ - 20^\circ = 70^\circ$. Since G is the midpoint of EF, $\angle CGF = \angle EGF = 70^\circ$. In triangle EOG, since G is the midpoint of EF, $\angle EOD = 2\angle EGF = 2 \times 70^\circ = 140^\circ$. Therefore, the answer is (D). The final answer is 10° .	
Verify output from MM-Verifier	
Let's verify the solution step by step:	
Understanding the Problem:	
1. Given: $\angle DCF = 20^\circ$, and G is the midpoint of EF.	
2. We need to find $\angle EOD$.	
Solution Steps Verification:	
1. Step 1: Since G is the midpoint of EF, triangle OEF is isosceles with $OE = OF$.	
2. Step 2: In circle O, $\angle DCF = 20^\circ$, which means $\angle DOE$ is a central angle subtending the same arc as $\angle DCF$.	
3. Step 3: Therefore, $\angle DOE = 2 \times \angle DCF = 2 \times 20^\circ = 40^\circ$.	
Conclusion:	
1. The solution correctly finds $\angle DOE = 40^\circ$.	
2. Since $\angle EOD$ is the same angle as $\angle DOE$, it should also be 40° .	
3. The final answer given is 10° , which is incorrect.	
Verification: Is the answer correct (Yes/No)? No.	✓
Verify output from Qwen2-VL-72B-Instruct	
Yes	✗

Figure 5: We present a case of MM-Verifier. We can see MM-Verifier correctly verify the answer with Long COT while Qwen2-VL-72B-Instruct failed to.

accurate responses. This iterative verification-refinement process significantly improved reasoning precision and robustness.

As shown in Table 4, our approach outperforms both open-source and closed-source MLLMs across multiple benchmarks. On the MATHVISTA dataset, our method achieves an overall accuracy of **65.3**, surpassing human performance

(**60.3**) and even GPT-4o (**63.8**). Similarly, in the MATHVERSE benchmark, our method consistently achieves strong performance with an overall score of **25.7**, outperforming strong baselines like Math-LLaVA-13B (**22.9**) and LLaVA-OneVision (**20.7**).

These results demonstrate the robustness of MM-Reasoner and MM-Verifier in improving mathematical reasoning across diverse tasks.

4.5 Case Study

Figure 5 presents a case study of MM-Verifier, demonstrating its ability to successfully identify logical errors in the reasoning process through step-by-step verification. In contrast, Qwen2-VL-72B-Instruct fails to provide a step-by-step reasoning trajectory, leading to an error in detecting these mistakes. This case underscores the superior analytical capabilities of MM-Verifier.

5 Conclusion

In this paper, we propose two data synthesis methods: the first generates long COT verification data, while the second synthesizes long COT inference data more efficiently. We use these synthetic data to train the MM-Verifier and MM-Reasoner. Our MM-Verifier not only outperforms larger models on the MathCheck benchmark but also demonstrates superior performance against larger models on benchmarks such as MathVista and MathVerse. Addi-

tionally, the MM-Reasoner exhibits strong scalability, with performance improving as the data size increases. Furthermore, the combination of MM-Verifier and MM-Reasoner achieves impressive results on the MathVista benchmark, surpassing even GPT-4o. These findings confirm the effectiveness of MM-Verifier and MM-Reasoner in enhancing multimodal reasoning tasks and lay the foundation for future advancements in this domain.

6 Limitations

Due to limited funding and computational resources, we were unable to scale our MM-Verifier and MM-Reasoner to Qwen2-VL-72B. Additionally, our scalability tests were restricted to datasets of fewer than 100K samples. We plan to conduct further experiments as soon as additional computational resources become available.

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A Performance of Naive MCTS in Multimodal Reasoning

In NLP-based mathematical reasoning tasks, given a question, the Monte Carlo Tree Search (MCTS) algorithm iteratively refines its responses through continuous self-reflection, ultimately converging to an optimized answer (Zhang et al., 2024a). Experimental results demonstrate that this search-based approach yields strong performance. Naturally, we hypothesized that applying the same methodology to multimodal tasks would lead to similar improvements. However, as shown in Table 6, we compare the performance of MCTS with Majority Voting and our MM-Verifier on the MathVista benchmark. We can see simple MCTS failed to improve reasoning performance. We attribute this to the significantly higher prevalence of hallucinations in multimodal models compared to their text-only counterparts. Therefore inspired by (Wang et al., 2024a; Sun et al., 2024) we design a simulation-based tree search.

B Detailed Performance of MM-Verifier on Sub-tasks of MathVista

Table 5 presents a comparative evaluation of Qwen2-VL-Instruct-7B, LLaMA-3.2-11B-Vision, and our proposed MM-Reasoner on the MathVista benchmark. The models were assessed across multiple reasoning-intensive sub-tasks, including GPS, MWP, FQA, TQA, and VQA.

In the Sample 4 evaluation, MM-Reasoner with the MM-Verifier (Stage 2) achieved an overall accuracy of 61.5, surpassing Qwen2-VL’s 59.8 and LLaMA-3.2-11B-Vision’s 50.4. This trend persists across Sample 8 and Sample 12, where MM-Reasoner obtained 65.3 and 65.7 respectively, further establishing its robustness in complex multimodal reasoning tasks.

Furthermore, the MM-Verifier mechanism contributes significantly to the accuracy gains. Across all models, the transition from Majority Voting to MM-Verifier (Stage 2) consistently improves performance, underscoring the importance of verification-enhanced reasoning. In particular, MM-Reasoner benefits the most from this verification process, indicating that it effectively integrates verification feedback into its reasoning pipeline. Overall, these results validate the effectiveness of MM-Verify as a strong algorithm.

Input <question, solution>	
Q: what is the total volume of the measuring cup? A: Let's think step by step. The answer is 1000.	
Verify output from MM-Verifier	
To verify the solution, let's analyze the given image and the solution step by step. 1. Identify the Measuring Cup: - The image shows a transparent measuring cup with markings for volume in milliliters (mL) and grams (g). 2. Observe the Markings: - The visible markings on the cup include the numbers 0, 200, 400, 600, 800, and 1000. 3. Check the Solution: - The solution states that the total volume is 1000 g, which corresponds to 1000 mL based on the markings on the cup. 4. Expected Answer: - The expected answer is 1000 mL, which matches the solution. 5. Conclusion: - The solution correctly identifies the total volume of the measuring cup as 1000 mL. Verification: Is the answer correct (Yes/No)? Yes	
Verify output from Qwen2-VL-72B-Instruct	
No	

Figure 6: We present case of MM-Verifier. We can see MM-Verifier correctly verify the answer with CoT while Qwen2-VL-72B-Instruct failed to.

C Prompts

This paper primarily focuses on three key prompts: (1) the prompt for generating Verify data (Figure 7), (2) the prompt (Figure 8) for distilling the QwQ-32B-Preview model after converting multimodal data into a textual format, and (3) the prompt for extracting answer data (Figure 9).

D More Case Study

As shown in Figure 6, our MM-Verifier is capable of verifying the correctness of a solution that provides only a simple answer by leveraging CoT approach. In contrast, Qwen2-VL-72B-Instruct incorrectly classifies the solution as correct, highlighting its limitations in reasoning-based verification.

Table 5: We compare sub-task’s performance of Qwen2-VL-Instruct-7B, LLaMA-3.2-11B-Vision with our MM-Reasoner on the MathVista benchmark. GPS: geometry problem solving; MWP: math word problem; FQA: figure question answering; TQA: textbook question answering; VQA: visual question answering.

Model	Method	GPS	MWP	FQA	TQA	VQA	ALL
<i>Sample 4</i>							
Qwen2-VL	Majority Voting	44.7	62.9	63.6	58.2	54.7	57.1
	Qwen2-VL-7B as Judgement	51.4	63.4	62.8	53.8	55.9	57.9
	MM-Verifier(Stage1)	51.0	65.1	62.5	63.3	52.0	58.8
	MM-Verifier(Stage2)	51.9	66.7	64.7	60.8	53.6	59.8
LLaMA-3.2-11B-Vision	Majority Voting	38.0	48.9	45.7	55.7	39.7	45.2
	Qwen2-VL-7B as Judgement	39.9	51.6	48.7	52.5	38.0	41.7
	MM-Verifier(Stage1)	42.3	54.8	51.7	57.0	45.3	50.0
	MM-Verifier(Stage2)	46.2	53.8	53.2	56.3	42.5	50.4
MM-Reasoner	Majority Voting	57.2	62.4	65.4	57.6	51.4	59.4
	Qwen2-VL-7B as Judgement	45.7	59.1	56.1	55.7	48.6	53.1
	MM-Verifier(Stage1)	51.0	65.1	65.1	61.4	54.2	59.6
	MM-Verifier(Stage2)	58.7	66.1	66.2	63.9	50.8	61.5
<i>Sample 8</i>							
Qwen2-VL	Majority Voting	55.8	65.1	65.8	62.0	55.3	61.1
	Qwen2-VL-7B as Judgement	49.0	60.2	59.1	52.5	49.7	54.5
	MM-Verifier(Stage1)	53.8	69.9	66.2	63.9	53.1	61.6
	MM-Verifier(Stage2)	52.4	70.4	69.5	62.7	55.3	62.5
LLaMA-3.2-11B-Vision	Majority Voting	43.8	52.7	50.6	57.0	38.0	48.3
	Qwen2-VL-7B as Judgement	39.9	51.6	48.7	52.5	38.0	46.1
	MM-Verifier(Stage1)	45.2	54.3	54.6	62.0	41.3	51.4
	MM-Verifier(Stage2)	44.7	55.4	55.0	63.9	42.5	52.1
MM-Reasoner	Majority Voting	57.2	66.7	66.2	64.6	55.3	62.2
	Qwen2-VL-7B as Judgement	47.6	52.2	59.1	56.3	51.4	53.6
	MM-Verifier(Stage1)	56.7	72.0	66.2	68.4	55.7	63.8
	MM-Verifier(Stage2)	60.0	73.1	68.8	67.7	55.9	65.3
<i>Sample 12</i>							
Qwen2-VL	Majority Voting	62.5	67.7	65.4	62.7	54.7	62.9
	Qwen2-VL-7B as Judgement	49.5	60.8	59.5	51.3	48.6	54.4
	MM-Verifier(Stage1)	56.7	69.9	69.9	65.2	54.7	63.7
	MM-Verifier(Stage2)	58.7	67.7	69.5	65.8	57.0	64.1
LLaMA-3.2-11B-Vision	Majority Voting	47.1	54.3	54.3	61.4	39.7	51.3
	Qwen2-VL-7B as Judgement	42.3	45.2	50.6	54.4	34.1	45.5
	MM-Verifier(Stage1)	50.5	57.5	58.0	65.8	43.6	55.0
	MM-Verifier(Stage2)	55.8	56.5	59.5	62.7	44.1	55.9
MM-Reasoner	Majority Voting	65.2	68.0	68.2	64.4	55.3	64.6
	Qwen2-VL-7B as Judgement	46.6	61.3	59.5	56.3	52.5	55.4
	MM-Verifier(Stage1)	61.1	72.0	70.3	64.6	53.1	64.8
	MM-Verifier(Stage2)	62.0	71.5	69.9	67.1	53.6	65.3

Table 6: Comparison of the performance of MCTS with Majority Voting and our MM-Verify on the MathVista benchmark.

Base Model	# size	Method	GPS	MWP	FQA	TQA	VQA	ALL
llama-3.2-11B-Vision	11B	MCTS	42.8	16.1	33.8	48.1	40.2	35.8
		Majority Voting	38.0	48.9	45.7	55.7	39.7	45.2
		MM-Verify	46.2	53.8	53.2	56.3	42.5	50.4
Qwen2-VL	7B	MCTS	53.4	47.3	51.7	55.7	43.6	50.4
		Majority Voting	44.7	62.9	63.6	58.2	54.7	57.1
		MM-Verify	51.9	66.7	64.7	60.8	53.6	59.8
LLaVA-OneVision	7B	MCTS	72.1	56.5	44.6	58.9	45.3	54.9
		Majority Voting	66.8	54.8	39.4	57.6	40.8	51.1
		MM-Verify	66.8	63.4	47.6	58.2	43.0	55.4

Verify Label Prompt
Solve the math problems and provide step-by-step solutions, ending with "The answer is [Insert Final Answer Here]". When asked "Verification: Is the answer correct (Yes/No)?", respond with " Yes" or " No" based on the answer's correctness. When asked "Verification: Let's verify step by step.", verify every step of the solution and conclude with "Verification: Is the answer correct (Yes/No)?" followed by " Yes" or " No". Q: {data['question']} A: Let's think step by step. {data['solution']}

Figure 7: Verify Label Prompt. This is the prompt we built with reference to Zhang et al. (2024c).

QwQ-32B-Preview Distill Prompt
Your role as an assistant involves thoroughly exploring questions through a systematic long thinking process before providing the final precise and accurate solutions. This requires engaging in a comprehensive cycle of analysis, summarizing, exploration, reassessment, reflection, backtracing, and iteration to develop well-considered thinking process. Please structure your response into two main sections: Thought and Solution. In the Thought section, detail your reasoning process using the specified format: <pre> """ < begin_of_thought > {thought with steps separated with "\n\n"} < end_of_thought > """ </pre> Each step should include detailed considerations such as analysing questions, summarizing relevant findings, brainstorming new ideas, verifying the accuracy of the current steps, refining any errors, and revisiting previous steps. In the Solution section, based on various attempts, explorations, and reflections from the Thought section, systematically present the final solution that you deem correct. The solution should remain a logical, accurate, concise expression style and detail necessary step needed to reach the conclusion, formatted as follows: <pre> """ < begin_of_solution > {final formatted, precise, and clear solution} < end_of_solution > """ </pre> Now, try to solve the following question through the above guidelines:

Figure 8: QwQ-32B-Preview Distill Prompt. This is the prompt we built with reference to Min et al. (2024).

Answer Extract Prompt
Hint: Please answer the question requiring an integer answer and provide the final value, e.g., 1, 2, 3, at the end. Question: Which number is missing? Model response: The answer is 14. <Extracted answer>: 14 ...
Hint: Please answer the question requiring a Python list as an answer and provide the final list, e.g., [1, 2, 3], [1.2, 1.3, 1.4], at the end. Question: Between which two years does the line graph saw its maximum peak? Model response: The line graph saw its maximum peak between 2007 and 2008. <Extracted answer>: [2007, 2008]
Hint: Please answer the question and provide the correct option letter, e.g., A, B, C, D, at the end. Question: What fraction of the shape is blue?\nChoices:\n(A) 3/11\n(B) 8/11\n(C) 6/11\n(D) 3/5 Model response: The correct answer is (B) 8/11. <Extracted answer>: B
Please extract the following <Extracted answer> referencing above example. Do not output any other information. Question: {data['question']} Model response: {data['response']} Extracted answer:

Figure 9: Answer Extract Prompt